Math 105L In Class 1 Practice

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Crash Course in Trig P1 and P2

For each of questions 1-10, draw a picture first, label it well, then compute. The pictures will be part of the grading on the test!

1. A drone takes off from level ground and rises at a constant angle of 42°. After traveling 2,500 feet along its path, how high above the ground is the drone?

2. A wheelchair ramp is built at an angle of 6° with the horizontal. If the platform is 3 feet above ground level, how long is the ramp?

3. A 20-foot ladder leans against a building, making a 55° angle with the ground. How high up the wall does the ladder reach?

4. An airplane descends at an angle of 12°. If its altitude is 7,200 feet, how far horizontally is it from the airport?

5. A surveyor stands 60 meters from the base of a cliff. The angle of elevation is 28°. Find the height of the cliff.

6. A rope 15 meters long is tied to the top of a pole. The rope makes a 70° angle with the ground. How tall is the pole?

7. Two satellites are each 9,000 km from Earth and subtend an angle of 68°. Estimate the distance between them.

   Note: 'subtend an angle' means the angle between the lines of sight from Earth to the satellites here. There was a similar problem on the worksheet!

8. From the top of a lighthouse 20 meters above sea level, the angle of depression to a boat is 15°. Find the distance to the boat.

9. Angles of elevation to the top of a tower are 38° and 22° from points 25 meters apart. Find the height of the tower.

10. From a cliff 40 meters high, the angle of depression to a buoy is 18°. How far is the buoy from the base?

11. Let the angle between the positive x-axis and the line $y = -10x$ be θ. Find the exact value of $\tan\theta$. What are all possible pairs of values of $\sin\theta$ and $\cos\theta$?

12. Let the angle between the positive x-axis and the line $y = \frac{1}{2}x$ be θ. If the terminal side of θ lies in the third quadrant, find the exact values of $\csc\theta$, $\sec\theta$, and $\cot\theta$.

13. A line makes an angle of θ with the positive x-axis. Suppose $\sec\theta = -\frac{5}{3}$ and $\sin\theta < 0$. Find the equation of the line and calculate the value of $\cot\theta$.

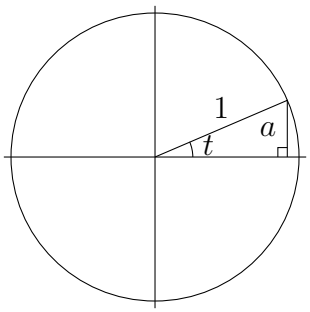
14. A geostationary orbit is one of altitude (distance above the surface of the earth) $22,236$ miles. Given that the radius of the Earth is $3,959$ miles, find the radian measure of an arc of length 500 miles along its orbit. Write your answer to two significant figures.

15. A circular race track has a radius of $1,200$ feet. A car starts at the point $(1200, 0)$ and travels counter-clockwise along the track for an arc length of $1,600$ feet.

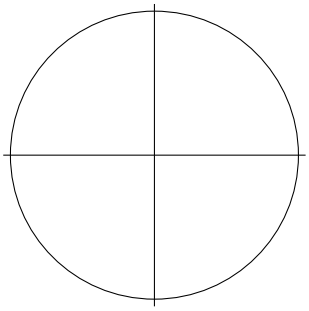
    (a) Find the radian measure of the angle θ, from the horizontal to the car's final position. Then convert your answer to degrees to one decimal place.

    (b) Find the coordinates (x, y) of the car's position to two decimal places.

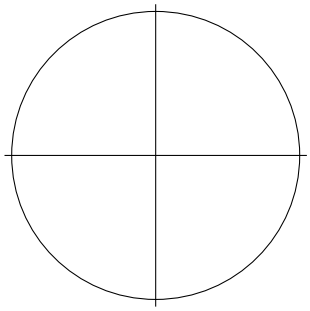
16. Let t be an angle in the first quadrant, as pictured below. For each of these, draw the given angle and a corresponding triangle congruent to the one given. Then use that to evaluate the given expression in terms of a.



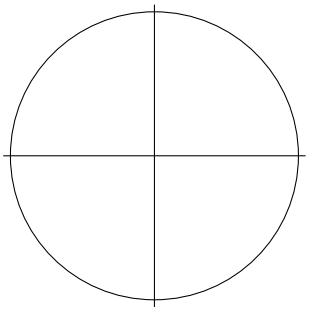
(a) $\cos(\pi + t) =$ _____



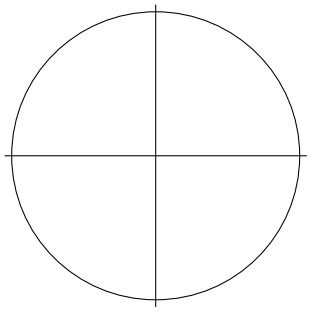
(c) $\tan(\pi - t) =$ _____



(b) $\sin\left(\frac{\pi}{2} + t\right) =$ _____



(d) $\sec\left(\frac{3\pi}{2} + t\right) =$ ____



17. For what values of θ between $0°$ and $360°$ are each of the following true? Explain your answers with words and/or drawings.

    (a) $\tan(\theta) < 0$

    (b) $\sec(\theta) \geq 2$

    (c) $\sin(\theta) \leq -\frac{\sqrt{2}}{2}$

Trig Identities and Limits

1. For each of the following, determine the smallest value $P > 0$ such that the statement is true for all θ in the domain. Explain your reasoning using the unit circle or the definition of the functions.

   (a) $\sin(\theta + P) = \sin(\theta)$ and $\cos(\theta + P) = \cos(\theta)$

   (b) $\tan(\theta + P) = \tan(\theta)$

   (c) Summarize the periods of all six trigonometric functions.

2. For each of the six trigonometric functions, determine the locations of their vertical asymptotes (VAs) between $0°$ and $360°$. Explain your reasoning by identifying which coordinate (x or y) in the denominator causes the function to be undefined.

   (a) Which functions have no vertical asymptotes?

   (b) At what values of θ do $\tan\theta$ and $\sec\theta$ have vertical asymptotes?

   (c) At what values of θ do $\cot\theta$ and $\csc\theta$ have vertical asymptotes?

3. Use the sum and difference formulas below to find the exact values of the given expressions.

$$\begin{aligned}\sin(A+B) &= \sin A\cos B + \cos A\sin B\\ \sin(A-B) &= \sin A\cos B - \cos A\sin B\\ \cos(A+B) &= \cos A\cos B - \sin A\sin B\\ \cos(A-B) &= \cos A\cos B + \sin A\sin B\end{aligned}$$

   (a) Find the exact value of $\sin(15°)$.

   (b) Find the exact value of $\cos(15°)$.

   (c) Find the exact value of $\sin(75°)$.

   (d) Find the exact value of $\cos(75°)$.

   (e) Using your answers for $\sin(15°)$ and $\cos(15°)$, find the exact value of $\tan(15°)$.

   (f) Using your answers for $\sin(75°)$ and $\cos(75°)$, find the exact value of $\tan(75°)$.

4. Using the facts that $\sin(2A) = 2\sin A\cos A$ and $\cos(2A) = \cos^2 A - \sin^2 A$, and the angle addition formulas above, find formulas for the following in terms of $\sin A$ and/or $\cos A$:

   (a) $\sin(3A)$

   (b) $\sin(4A)$

5. Compute the following limits (you may *not* use L'Hôpital's rule here):

   (a) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x}$ (Hint: Multiply the numerator and denominator by $1+\cos x$.)

   (b) $\lim\limits_{x\to 0}\dfrac{\tan x}{x}$ (Hint: Rewrite $\tan x$ in terms of $\sin x$ and $\cos x$.)

(c) $\lim_{x\to 0} \dfrac{\sin(2x)}{x\cos(2x)}$ (Hint: Use $\sin(2x) = 2\sin x \cos x$.)

(d) $\lim_{x\to\infty} \dfrac{\cos^2 x}{x^2}$ (Hint: What are the lower and upper bounds for the numerator?)

6. Prove the following identities. In each case, you must start from the expression on the left and arrive at the right. You may not manipulate both sides. Try rewriting each in terms of $\sin x$ and $\cos x$.

(a) $\frac{\sec x}{\csc x} = \tan x$

(b) $\sec\theta - \cos\theta = \sin\theta\tan\theta$

(c) $\cos x(\sec x - \cos x) = \sin^2 x$

(d) $\tan^2 x \cos^2 x + \cos^2 x = 1$

7. Prove the following identities. In each case, you must start from the expression on the left and arrive at the right. You may not manipulate both sides.

You will need the identities:

$$\sin(2A) = 2\sin A\cos B$$
$$\cos(2A) = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

(a) $\dfrac{\sin(2x)}{\sin x} - \dfrac{\cos(2x)}{\cos x} = \sec x$

(b) $(\sin x + \cos x)^2 = 1 + \sin(2x)$

(c) $\dfrac{1-\cos(2x)}{\sin(2x)} = \tan x$

(d) $\cos^4 x - \sin^4 x = \cos(2x)$ (Hint: This is a difference of squares. Challenge question! Not on the test, but worth a try if you've done all the rest.)

Differentiating Trig Functions

1. What function is closely approximated by $\frac{\tan(x+0.00001)-\tan x}{0.00001}$? Explain.

2. Find the equation of the tangent line for each of the following functions at the given point.

   (a) $f(x) = \sin x$ at $x = \pi$

   (b) $f(x) = \tan x$ at $x = \frac{\pi}{4}$

   (c) $f(x) = \cos(2x)$ at $x = \frac{\pi}{6}$

   (d) $f(x) = x \sin x$ at $x = \frac{\pi}{2}$

   (e) $f(x) = x^2 \cos x$ at $x = \pi$

3. For each of the following functions, find the linear approximation $L(x)$ at the indicated point, then use that approximation to estimate the given value. Provide exact values using fractions, π, and roots.

   (a) $f(x) = \sin(2x)$ at $x = \frac{\pi}{3}$. Estimate $\sin(\frac{7\pi}{18})$.

   (b) $h(x) = \tan x$ at $x = \frac{\pi}{4}$. Estimate $\tan(\frac{5\pi}{24})$.

   (c) $f(x) = \cos^2 x$ at $x = \frac{\pi}{6}$. Estimate $\cos^2(\frac{\pi}{5})$.

   (d) $g(x) = \csc x$ at $x = \frac{\pi}{2}$. Estimate $\csc(\frac{11\pi}{20})$.

   (e) $f(x) = \cos(x + \frac{\pi}{4})$ at $x = 0$. Estimate $\cos(\frac{\pi}{4} + \frac{1}{10})$.

4. For each function, find all critical points in the interval $[0, 2\pi]$. After differentiating, you will need the double angle formulas:

$$\sin(2A) = 2 \sin A \cos B$$
$$\cos(2A) = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1$$

   You will also need to factor or use the quadratic formula.

   (a) $y = 2 \sin x + \cos(2x)$

   (b) $y = \cos(2x) - 2 \cos x$

   (c) $y = \cos(2x) + 2 \sin x$

   (d) $y = \sin x + \frac{1}{2} \cos(2x)$

   (e) $y = \sin(2x) - 3 \sin x$ (Note: This one requires inverse trig functions.)

Inverse Trig Functions

For Q1-5, solve the given equation on the given domain. In each case, draw the graph of the respective base function (e.g. $\cos x$ for Q3), and indicate solutions on it using horizontal lines. Also draw the unit circles for each, showing all solutions.

1. $2\sin x + \sqrt{2} = 0$ for $0 \leq x \leq 2\pi$

2. $3\tan x - 1 = 2$ for $0 \leq x \leq 2\pi$

3. $4\cos^2 x - 1 = 2$ for $0 \leq x \leq 2\pi$

4. $2\sin^2 x - \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$

5. $2\cos^2 x + 5\cos x + 1 = 0$ for $0 \leq x \leq 2\pi$

6. Explain why $f(x) = \sin x$ is considered invertible on the domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$ but not on the domain $[-2\pi, 2\pi]$.

7. Describe the geometric relationship between the graph of a trigonometric function (on its restricted domain) and the graph of its inverse.

8. For what specific interval of x is the statement $\cos^{-1}(\cos x) = x$ true? Why is it not true for all real numbers?

9. Complete the following conceptual mappings:

   - $\sin^{-1}(x)$ takes a _____ and returns an _____.
   - $\tan^{-1}(x)$ takes a _____ and returns an _____.

10. Describe what happens to the vertical asymptotes of $y = \tan x$ when the function is inverted to $y = \arctan x$. What is $\lim_{x \to -\infty} \arctan(x)$?

Differentiating Inverse Trig Functions

1. To invert $\cot x$, we restrict it to the domain $0 < x < \pi$. We call its inverse arccot x.

   Find the derivative of $y = \text{arccot}(x)$ using the derivative of $\cot(x)$ and implicit differentiation. You will need to find an identity relating $\csc^2 y$ to $\cot^2 y$ along the way.

2. Find the derivative of $f(x) = x^2 \arctan(x)$.

3. Find the derivative of $g(x) = \arcsin(e^x)$.

4. Find the derivative of $h(x) = \frac{\arccos(x)}{x+1}$.

5. Find the derivative of $y = \arctan(\sqrt{x})$.

6. Find the derivative of $f(x) = \ln(\arcsin(x))$.

7. Use linear approximation to estimate the value of $\arcsin(0.51)$.

8. Use the tangent line to $f(x) = \arctan(x)$ at $x = 1$ to approximate $\arctan(1.05)$.

9. Use linear approximation to estimate the value of $f(0.1)$ for the composite function $f(x) = \arctan(1 - 2x)$.

10. (a) What is $\tan(\arctan x)$?

    (b) Given the above, what should $\frac{d}{dx} \tan(\arctan x)$ be?

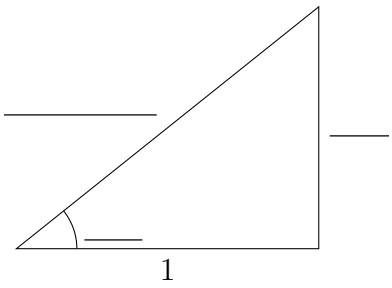
    (c) Use the chain rule instead to differentiate $\tan(\arctan x)$.

    (d) What must be true about your last two answers?

    (e) Let $y = \arctan x$. What kind of object is y? Label it on the triangle below.

    (f) If the bottom length of the triangle has length 1, label x on the triangle as well.

    (g) Compute the length of the third side. Then, use the triangle to find $\sec(\arctan x)$. How does this fit in with your answer to (d)?



$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

Trig Related Rates

1. **The Rotating Searchlight**
   A searchlight 20 meters from a straight wall rotates at a constant rate of 0.5 rad/s. At what rate is the beam moving along the wall when the beam makes an angle of $\pi/4$ with the line perpendicular to the wall?

2. **The Rising Balloon**
   A hot air balloon rises vertically from a level field at a rate of 3 m/s. An observer is 100 meters from the lift-off point. At what rate is the angle of elevation changing when the balloon is 100 meters high?

3. **Telescope Tracking**
   A plane flies horizontally at 600 mph at an altitude of 5 miles. Find the rate of change of the telescope angle from the ground, θ, when the horizontal distance from the telescope is 12 miles.

4. **Two Planes Crossing**
   Plane A flies north at 400 km/h toward an airport. Plane B flies east at 300 km/h toward the same airport. Let θ be the bearing of A from B. Find $d\theta/dt$ when A is 4 km and B is 3 km from the airport.

   Hint: you have three variables here – an angle and two distances. You'll need the quotient rule along the way.

5. **Rocket Tracking**
   A camera 3000 ft from a launch pad tracks a rocket with height $s = 50t^2$. Find the rate of change of the angle θ when $t = 10$ s.

6. **Sliding Ladder**
   A 10-ft ladder leans against a wall. The bottom slides away at 2 ft/s. How fast is the angle between the ladder and the floor changing when the bottom is 6 ft from the wall?

7. **The Ferris Wheel**
   A Ferris wheel (radius 10m) rotates at 1 rad/min. How fast is a passenger rising when they are 8m above the center's horizontal line?

8. **The Moving Shadow**
   A man 2 meters tall walks away from a 6-meter lamppost at 1.5 m/s. Let θ be the angle of the light beam hitting the ground. How fast is θ changing when the man is 8 meters from the post?

   Hint: This one is harder, and involves similar triangles. Let the distance of the man from the post be x_m and the distance of the tip of the shadow from the post be x. You'll need to find a relationship between these two!